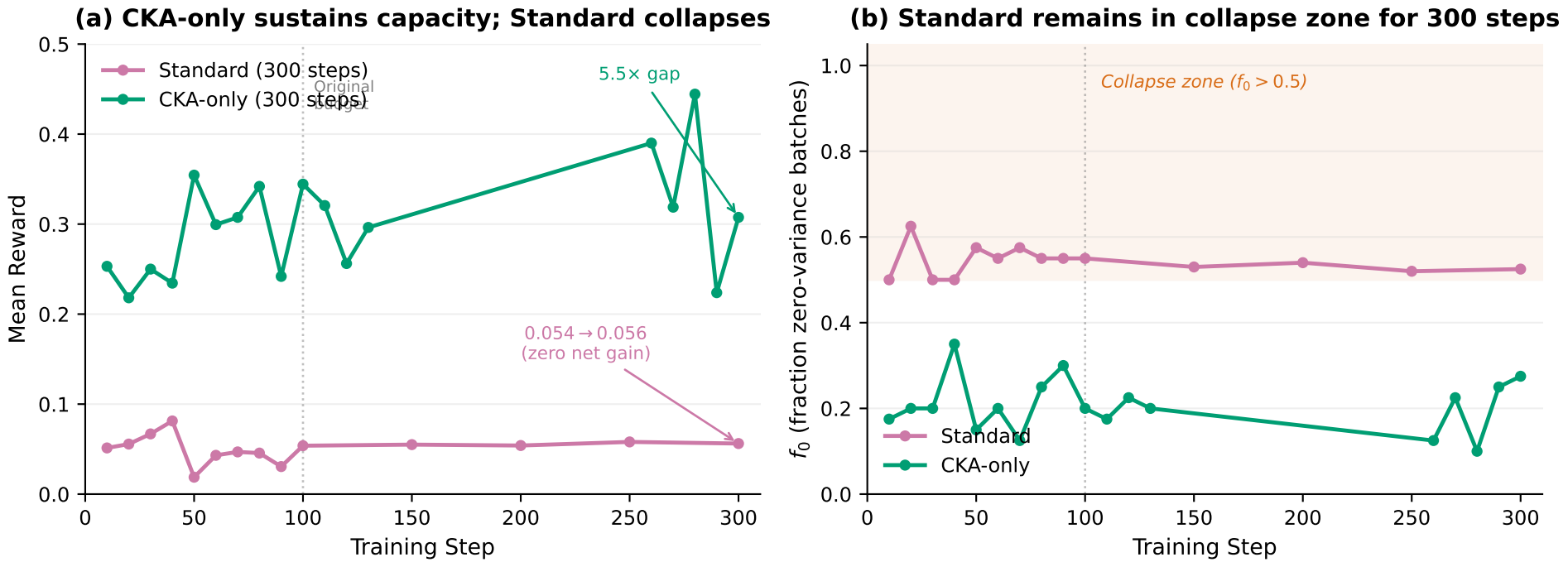
(a) CKA-only sustains capacity; Standard collapses
Standard (300 steps)
CKA-only (300 steps)
Original budget
5.5× gap
0.054 → 0.056
(zero net gain)
Mean Reward
Training Step
(b) Standard remains in collapse zone for 300 steps
Collapse zone ($f_0 > 0.5$)
f_0 (fraction zero-variance batches)
Standard
CKA-only
Training Step